

In [7]: `import numpy as np`

```
def f(x):
    return 4 * x**4 + 3 * x**3 - 2 * x**2 + 5 * x - 7

a, b = 0, 2
n = 4
h = (b - a) / n
x = np.linspace(a, b, n + 1)
y = f(x)

integral_approx = (h / 3) * (f(a) + 4 * np.sum(f(x[1:-1:2])) + 2 * np.sum(f(x[2:-2:2])))
print(f"Approximate integral (Simpson's 1/3 Rule): {integral_approx:.4f}")

def F(x):
    return (4 * x**5 / 5) + (3 * x**4 / 4) - (2 * x**3 / 3) + (5 * x**2 / 2) - 7 * x

exact = F(b) - F(a)
print(f"Exact integral: {exact:.4f}")

error = abs(integral_approx - exact) / exact * 100
print(f"Percentage error: {error:.4f}%")
```

Approximate integral (Simpson's 1/3 Rule): 28.3333

Exact integral: 28.2667

Percentage error: 0.2358%

In []: